Phase 1: Modelling, Equilibria and Linearization of the Acrobot

Bhargav Sreekar Kaushik Divijat
2025102061 2025102074 2025102071 2025102040

Control Systems Project — Phase 1 Report — October 25, 2026

***Abstract*—We derive and audit the nonlinear model of the Acrobot, a two-link planar robot actuated only at the elbow. The full equilibrium set is found in closed form: four unforced postures and two one-parameter families of forced equilibria, all of which can be held within the $20\,\mathrm{N\,m}$ motor limit. The dynamics are linearized analytically at every equilibrium, and state-space models and transfer functions $\tau \to q_1$, $\tau \to q_2$ are given at the four unforced postures. All linear models are controllable, and observable from either joint angle alone. At the upright posture, however, two unstable poles combine with non-minimum-phase zeros to severely limit single-angle feedback. The model is checked against nonlinear simulation and energy conservation.**

I. Introduction

The Acrobot is a planar double pendulum with a motor at the elbow and a passive shoulder [1,2]. It has two degrees of freedom but one input, so the shoulder can only be moved through coupling. The project aims to swing it up and balance it upright with $|\tau| \le \tau_{\max} = 20\,\mathrm{N\,m}$. Phase 1 provides the verified model, all equilibria, and the linear models and transfer functions at those equilibria.

II. Nonlinear Model

A. Coordinates and Parameters

The shoulder angle q_1 is measured counter-clockwise from the downward vertical, and the elbow angle q_2 is measured relative to link 1 (Fig. 1). The state is $x = [q_1, q_2, \dot{q}_1, \dot{q}_2]^T$, and the torque enters through $B = [0, 1]^T$. The links are uniform rods with $m_1 = m_2 = 1$ kg, $l_1 = 1$ m, $l_2 = 2$ m, $l_{c1} = 0.5$ m, $l_{c2} = 1$ m, $I_1 = 0.083$ kg m^2 and $I_2 = 0.33$ kg m^2 (about the COM), and $g = 9.81$ m/s^2.

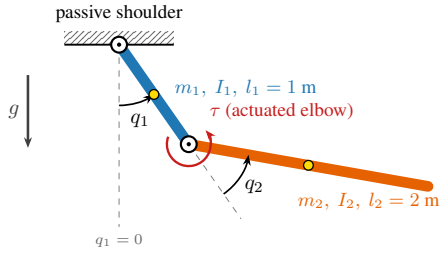


Fig. 1. Acrobot schematic (to scale); dots mark the centres of mass.

B. Lagrangian Derivation

Let $\varphi = q_1 + q_2$ be the absolute angle of link 2. The centres of mass are at $p_{c1} = l_{c1}[\sin q_1, -\cos q_1]^T$ and $p_{c2} = l_1[\sin q_1, -\cos q_1]^T + l_{c2}[\sin\varphi, -\cos\varphi]^T$, so $\|\dot{p}_{c2}\|^2 = l_1^2\dot{q}_1^2 + l_{c2}^2\dot{\varphi}^2 + 2l_1 l_{c2}\cos q_2\,\dot{q}_1\dot{\varphi}$. The kinetic energy is then a quadratic form,

$$T = \tfrac{1}{2}\textstyle\sum_i m_i\|\dot{p}_{ci}\|^2 + \tfrac{1}{2}I_1\dot{q}_1^2 + \tfrac{1}{2}I_2\dot{\varphi}^2 = \tfrac{1}{2}\dot{q}^T M(q)\dot{q}. \tag{1}$$

With the datum at the pivot, the potential energy is

$$\begin{aligned} U &= -g[(m_1 l_{c1} + m_2 l_1)\cos q_1 + m_2 l_{c2}\cos\varphi] \\ &= -g[1.5\cos q_1 + \cos\varphi]. \end{aligned} \tag{2}$$

The Euler–Lagrange equations for $L = T - U$ then give

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} = \tau_g(q) + B\tau, \qquad \tau_g = -\nabla_q U. \tag{3}$$

C. Model Matrices

Writing $c_2 = \cos q_2$ and $s_2 = \sin q_2$,

$$\begin{aligned} M_{11} &= I_1 + I_2 + m_1 l_{c1}^2 + m_2(l_1^2 + l_{c2}^2 + 2l_1 l_{c2}c_2) = 2.663 + 2c_2, \\ M_{12} &= M_{21} = I_2 + m_2(l_{c2}^2 + l_1 l_{c2}c_2) = 1.33 + c_2, \\ M_{22} &= I_2 + m_2 l_{c2}^2 = 1.33, \end{aligned} \tag{4}$$

$$C = s_2\begin{bmatrix} -2\dot{q}_2 & -\dot{q}_2 \\ \dot{q}_1 & 0 \end{bmatrix}, \quad \tau_g = -9.81\begin{bmatrix} 1.5\sin q_1 + \sin\varphi \\ \sin\varphi \end{bmatrix}, \tag{5}$$

where C carries the factor $m_2 l_1 l_{c2} = 1$. The state model $\dot{x} = f(x,\tau)$ used throughout stacks $\dot{q}$ and

$$\ddot{q} = M(q)^{-1}\big(\tau_g(q) + B\tau - C(q,\dot{q})\dot{q}\big), \tag{6}$$

D. Model Audit and Structural Properties

- *Handout.* With $l_{ci} = l_i/2$, (4)–(5) reproduce the handout's M, C and U term by term.
- *Invertibility.* $\det M = 1.7729 - \cos^2 q_2 \in [0.7729, 1.7729]$ and $M_{22} > 0$, so M is positive definite for every q.
- *Energy balance.* $\dot{q}^T(\tfrac{1}{2}\dot{M} - C)\dot{q} = 0$ for this C, so the total energy $E = T + U$ satisfies $\dot{E} = \dot{q}_2\tau$. Numerically, with $\tau = 0$, E drifts by only 6×10^{-9} (relative) over 20 s.
- *Rounded inertias.* Using $0.083, 0.33$ instead of $1/12, 1/3$ shifts the natural frequencies by at most 0.26%.

III. Equilibria

A. Equilibrium Conditions

Setting $\dot{q} = \ddot{q} = 0$ in (3) leaves $\tau_g(q^*) + B\tau^* = 0$:

$$\text{shoulder:} \quad 1.5\sin q_1^* + \sin\varphi^* = 0, \tag{7}$$

$$\text{elbow:} \quad \tau^* = 9.81\sin\varphi^* = -14.715\sin q_1^*\ \mathrm{N\,m}. \tag{8}$$

Eq. (7) says gravity exerts no torque about the passive shoulder, i.e. the combined centre of mass lies on the vertical through the pivot; (8) is the motor's holding torque.

B. Forced Equilibria ($\tau^* \neq 0$)

Solving (7) for φ^* gives two branches,

$$q_2^* = \begin{cases} \arcsin(-1.5\sin q_1^*) - q_1^*, \\ \pi - \arcsin(-1.5\sin q_1^*) - q_1^*, \end{cases} \tag{9}$$

which are real if and only if $|\sin q_1^*| \le 2/3$. This gives a *down family*, $q_1^* \in [-41.81^\circ, 41.81^\circ]$, containing E_1 and E_2, and an *up family*, $q_1^* \in [138.19^\circ, 221.81^\circ]$, containing E_3 and E_4. Within each family the two branches meet where $|\sin q_1^*| = 2/3$, i.e. $\varphi^* = \pm\pi/2$ (link 2 horizontal), so each family is a smooth closed curve (Fig. 2a). The holding torque (8) is largest at these fold points, $|\tau^*|_{\max} = m_2 g l_{c2} = 9.81\,\mathrm{N\,m}$, which is the full weight of link 2 at lever arm l_{c2}. Since $9.81 < 20\,\mathrm{N\,m}$, *every* equilibrium can be held without saturating the motor (Fig. 2b).

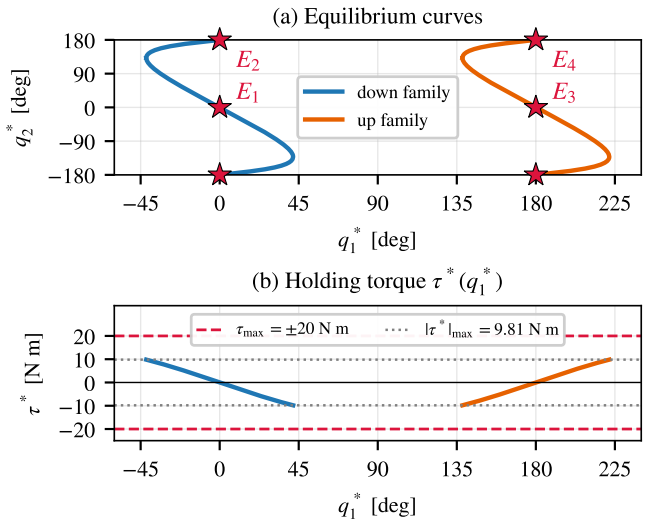


Fig. 2. (a) Equilibrium set in (q_1^*, q_2^*); red stars are the unforced equilibria ($q_2 = \pm 180°$ is the same posture). (b) Holding torque along both families.

C. Unforced Equilibria ($\tau^* = 0$)

With $\tau^* = 0$, (8) forces $\sin\varphi^* = 0$, and then (7) forces $\sin q_1^* = 0$. This gives the four postures in Table I; their energies follow from (2).

TABLE I
UNFORCED EQUILIBRIA

	Posture	(q_1^*, q_2^*)	U [J]	Remark
E_1	down–down	$(0, 0)$	-24.525	minimum of U
E_2	down–up	$(0, \pi)$	-4.905	link 2 folded up
E_3	up–up	$(\pi, 0)$	$+24.525$	upright target; max. of U
E_4	up–down	(π, π)	$+4.905$	link 2 hanging

IV. Linearization

A. Jacobians at a General Equilibrium

Let $\tilde{x} = x - x^*$ and $\tilde{\tau} = \tau - \tau^*$. Two observations simplify the Jacobian of (6). First, $C\dot{q}$ is quadratic in $\dot{q}$, so its Jacobian vanishes at $\dot{q}^* = 0$. Second, with $w = \tau_g + B\tau$, $\partial(M^{-1}w)/\partial q_j = (\partial M^{-1}/\partial q_j)w + M^{-1}\partial w/\partial q_j$, and $w = 0$ at an equilibrium. Hence, at *any* equilibrium, forced or not,

$$\dot{\tilde{x}} = A\tilde{x} + B_\ell \tilde{\tau}, \quad A = \begin{bmatrix} 0 & I_2 \\ P & 0 \end{bmatrix}, \quad B_\ell = \begin{bmatrix} 0 \\ r \end{bmatrix}, \tag{10}$$

where $P = M(q^*)^{-1}K(q^*)$, $r = M(q^*)^{-1}B$, and K is the gravity stiffness

$$K(q^*) = \frac{\partial \tau_g}{\partial q}\bigg|_{q^*} = -9.81 \begin{bmatrix} 1.5\cos q_1^* + \cos\varphi^* & \cos\varphi^* \\ \cos\varphi^* & \cos\varphi^* \end{bmatrix}. \tag{11}$$

B. Linear Models at the Unforced Equilibria

At all four postures $\cos q_2^* = \pm 1$, so $\det M = 0.7729$. Moving the shoulder from 0 to π leaves M unchanged but flips the sign of K, hence $P_{E3} = -P_{E1}$ and $P_{E4} = -P_{E2}$, with the same r. For E_1 and E_3, where $q_2^* = 0$ and $M = \left[\begin{smallmatrix} 4.663 & 2.33 \\ 2.33 & 1.33 \end{smallmatrix}\right]$,

$$P_{E1} = \begin{bmatrix} -12.6292 & 12.6926 \\ 14.7488 & -29.6119 \end{bmatrix}, \quad r_{E1,E3} = \begin{bmatrix} -3.0147 \\ 6.0332 \end{bmatrix}. \tag{12}$$

For E_2 and E_4, where $q_2^* = \pi$ and $M = \left[\begin{smallmatrix} 0.663 & 0.33 \\ 0.33 & 1.33 \end{smallmatrix}\right]$,

$$P_{E2} = \begin{bmatrix} -12.6292 & 12.6926 \\ 10.5095 & 4.2266 \end{bmatrix}, \quad r_{E2,E4} = \begin{bmatrix} -0.4270 \\ 0.8578 \end{bmatrix}. \tag{13}$$

C. Stability Along the Forced Families

Evaluating (10) along Fig. 2a, each branch keeps the number of unstable modes of the unforced equilibrium on it (0, 1, 2, 1 for the E_1, E_2, E_3, E_4 branches). It changes only at the folds, where $\cos\varphi^* = 0$ makes P singular.

V. Transfer Functions and System Properties

A. Transfer Functions

Laplace-transforming (10) gives $(s^2 I - P)\tilde{Q}(s) = r\,\tilde{\mathscr{T}}(s)$. Inverting the 2×2 matrix yields $G_i = \tilde{Q}_i/\tilde{\mathscr{T}}$:

$$\begin{aligned} G_1(s) &= \big[r_1 s^2 + (P_{12}r_2 - P_{22}r_1)\big]/\Delta(s), \\ G_2(s) &= \big[r_2 s^2 + (P_{21}r_1 - P_{11}r_2)\big]/\Delta(s), \end{aligned} \tag{14}$$

where $\Delta(s) = s^4 - \mathrm{tr}(P)\,s^2 + \det P$ is monic and shared by both outputs (values in Table II).

B. Structure of Poles and Zeros

- *Symmetry.* There is no damping, so (14) depends on s^2 only. Each mode is therefore either a center ($\pm j\omega$) or a saddle ($\pm\lambda$), and every pair of real zeros includes an RHP (non-minimum-phase) zero.
- *Gravity reversal.* Since $P_{E3} = -P_{E1}$, the map $s^2 \to -s^2$ turns the oscillation frequencies 2.240 and 6.101 rad/s at E_1 into divergence rates at E_3. The fastest mode doubles every $\ln 2/6.101 = 0.114$ s; E_2/E_4 are related likewise.
- *Meaning of the zeros.* With $q_2 \equiv 0$, row 1 of the linearized (3) is the locked-elbow pendulum $M_{11}\ddot{q}_1 = K_{11}q_1$, so the zeros of G_2 satisfy $s^2 = K_{11}/M_{11}$. Holding $q_1 \equiv 0$ with no shoulder torque needs $M_{12}\ddot{q}_2 = K_{12}q_2$, so the zeros of G_1 satisfy $s^2 = K_{12}/M_{12}$ (at E_1: -5.2595 and -4.2103, as in Table II).

C. Controllability and Observability

The Kalman matrix $\mathscr{C} = [B_\ell, AB_\ell, A^2B_\ell, A^3B_\ell]$ has rank 4 at E_1–E_4, with $\det\mathscr{C} = -364.15$ at $E_{1,3}$ and -184.90 at $E_{2,4}$. It also keeps rank 4 along both forced families, where $|\det\mathscr{C}| \geq 7.94$. Physically, the elbow torque always reaches the shoulder through the inertial coupling $M_{12} = 1.33 + \cos q_2 \in [0.33, 2.33]$, which never vanishes. The observability matrix has rank 4 at E_1–E_4 for $y = [q_1, q_2]^T$, and also for q_1 alone or q_2 alone.

D. Implications for Control at E_3

Observability from a single angle means a stabilizing dynamic output-feedback controller exists in principle. An RHP zero z and RHP poles p_i, however, force $M_s = \|S\|_\infty \geq \prod_i |z + p_i|/|z - p_i|$ for every stabilizing loop [3]. Feeding back q_1 ($z = 2.052$) gives $M_s \geq 45.9$, and feeding back q_2 ($z = 2.293$) gives $M_s \geq 187$; both are far above the usual target $M_s < 2$. Moreover, only one real RHP pole (6.101) lies between q_2's zero at 2.293 and the zero at infinity. By the parity interlacing property [4], any controller that stabilizes E_3 from q_2 alone must therefore be unstable itself. (At E_2 and E_4 the bounds are only 3.8 and 4.4.) Balancing E_3 therefore calls for full-state feedback.

TABLE II

TRANSFER FUNCTIONS $G_i(s) = N_i(s)/\Delta(s)$, POLES, ZEROS AND STABILITY AT THE UNFORCED EQUILIBRIA

	Denominator $\Delta(s)$	$N_1(s)$ $(\tau \to q_1)$	$N_2(s)$ $(\tau \to q_2)$	Poles	Zeros of G_1	Zeros of G_2	Type (RHP poles)
E_1	$s^4 + 42.2410s^2 + 186.7719$	$-3.0147(s^2 + 4.2103)$	$6.0332(s^2 + 5.2595)$	$\pm j2.2400,\ \pm j6.1011$	$\pm j2.0519$	$\pm j2.2934$	center–center (0)
E_2	$s^4 + 8.4025s^2 - 186.7719$	$-0.4270(s^2 - 29.7273)$	$0.8578(s^2 + 7.3982)$	$\pm 3.1775,\ \pm j4.3010$	± 5.4523	$\pm j2.7200$	saddle–center (1)
E_3	$s^4 - 42.2410s^2 + 186.7719$	$-3.0147(s^2 - 4.2103)$	$6.0332(s^2 - 5.2595)$	$\pm 2.2400,\ \pm 6.1011$	± 2.0519	± 2.2934	saddle–saddle (2)
E_4	$s^4 - 8.4025s^2 - 186.7719$	$-0.4270(s^2 + 29.7273)$	$0.8578(s^2 - 7.3982)$	$\pm 4.3010,\ \pm j3.1775$	$\pm j5.4523$	± 2.7200	saddle–center (1)

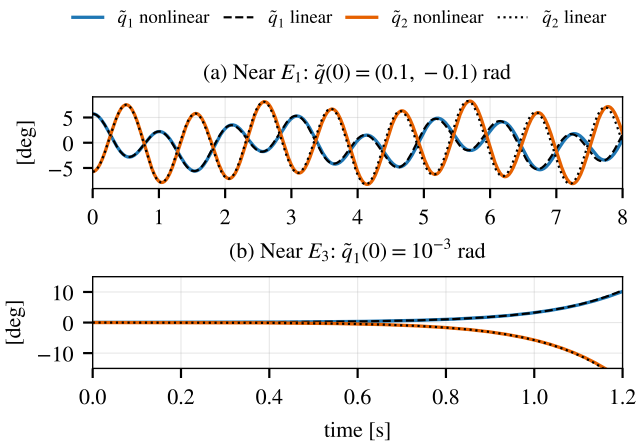


Fig. 3. Nonlinear (colour) vs. linearized (black) free responses (DOP853, rel. tol. 10^{-10}) about (a) E_1 and (b) E_3.

VI. VERIFICATION

- *Numerical values.* `Acrobot_Phase1_Main.m` derives A, B_ℓ symbolically (`jacobian`) and computes the transfer functions, poles, zeros, $\mathscr{C}$ and $\mathscr{O}$. An independent Python/NumPy recomputation agrees to four decimals.
- *Linear vs. nonlinear* (Fig. 3). Near E_1, starting from $\tilde{q} = (0.1, -0.1)$ rad, the maximum errors over 8 s are 1.0° (q_1) and 2.0° (q_2). Near E_3, from $\tilde{q}_1 = 10^{-3}$ rad, the error is 0.02% when $|\tilde{q}_1|$ reaches 1° ($t = 0.80$ s) and 0.37° at 10° ($t = 1.20$ s).

VII. CONCLUSION

The model is verified; all equilibria are holdable within $\tau_{\max}$; (10)–(11) linearize each; and at E_3, two unstable modes plus RHP zeros rule out practical single-angle feedback.

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